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LAB : 6

QUESTION : 1

1. Create a database called ‘vehicles’ and write a MongoDB query to select database as “vehicles”.

```
> use vehicles
switched to db vehicles
>
```

2. Write a MongoDB query to display all the databases.

```
> show dbs
admin     0.000GB
config    0.000GB
local     0.000GB
vehicles  0.000GB
>
```

3. Create a collection called ‘two_wheelers’. (use capping) and Create a collection called ‘Four_wheelers’.

```
> db.createCollection("two_wheelers" , {capped:true, size: 5000, max: 100})
{ "ok" : 1 }
> db.createCollection("four_wheelers")
{ "ok" : 1 }
>
```

4. Add 5 two-wheeler details to the collection named ‘two_wheelers’. Each document consists of following fields as bike_name, model (gear or gearless), category (100cc, 125cc, 150cc, 200cc), colors_available (red, black, blue, sport red etc) as array, manufacturer, performance (out of 10), timestamp (date and year release) and Price.

```
> db.two_wheelers.insertMany([{ bike_name: "Splendor", model: "gear", category: "100cc", colors_ava
ilable: ["black", "purple"], manufacturer: "Hero", performance: 8, timestamp: new Date("2023-01-10"
), p  { bike_name: "Activa", model: "gearless", category: "125cc", colors_available: ["white", "red
"],   { bike_name: "Pulsar", model: "gear", category: "150cc", colors_available: ["black", "sport r
ed"]  { bike_name: "Duke", model: "gear", category: "200cc", colors_available: ["orange", "white"],
 man  { bike_name: "Jupiter", model: "gearless", category: "110cc", colors_available: ["blue", "gre
y"], manufacturer: "TVS", performance: 4, timestamp: new Date("2023-08-12"), price: 78000 }e", "gre
... ])
{
        "acknowledged" : true,
        "insertedIds" : [
                ObjectId("69b0ebf28f1c10df3acef7eb"),
                ObjectId("69b0ebf28f1c10df3acef7ec"),
                ObjectId("69b0ebf28f1c10df3acef7ed"),
                ObjectId("69b0ebf28f1c10df3acef7ee"),
                ObjectId("69b0ebf28f1c10df3acef7ef")
        ]
}
> |
```

5. Add 5 four-wheeler details to the collection named ‘four_wheelers’. Each document consists of following fields as vehicle_name, model (commercial or own), category (car, lorry, bus, mini truck, heavy truck, containers), variants (vxi, zxi, petrol, diesel etc) as array, manufacturer, performance (out of 10), timestamp (date and year release) and price.

```
> db.four_wheelers.insertMany([
... { vehicle_name: "Swift", model: "own", category: "car", variants: ["vxi", "zxi"
, "petrol"], manufacturer: "Maruti", performance: 8, timestamp: new Date("2023-03-1
0"), price: 700000 },
...   { vehicle_name: "City", model: "own", category: "car", variants: ["petrol", "
diesel"], manufacturer: "Honda", performance: 9, timestamp: new Date("2022-09-15"),
 price: 1300000 },
...   { vehicle_name: "Eicher 10.5", model: "commercial", category: "mini truck", v
ariants: ["diesel"], manufacturer: "Eicher", performance: 7, timestamp: new Date("2
021-05-10"), price: 1500000 },
...   { vehicle_name: "Marcopolo", model: "commercial", category: "bus", variants:
["diesel"], manufacturer: "Tata", performance: 4, timestamp: new Date("2022-12-01")
, price: 2800000 },
...   { vehicle_name: "Thar", model: "own", category: "car", variants: ["lx", "dies
el", "petrol"], manufacturer: "Mahindra", performance: 9, timestamp: new Date("2023
-06-25"), price: 1600000 }
... ])
{
        "acknowledged" : true,
        "insertedIds" : [
                ObjectId("69b0ec758f1c10df3acef7f5"),
                ObjectId("69b0ec758f1c10df3acef7f6"),
                ObjectId("69b0ec758f1c10df3acef7f7"),
                ObjectId("69b0ec758f1c10df3acef7f8"),
                ObjectId("69b0ec758f1c10df3acef7f9")
        ]
}
>
```

6. Write a MongoDB query to display all documents available in two_wheelers and Four_wheelers.

```
> db.two_wheelers.find().pretty()
{
        "_id" : ObjectId("69b0eb708f1c10df3acef7e1"),
        "bike_name" : "Splendor",
        "model" : "gear",
        "category" : "100cc",
        "colors_available" : [
                "black",
                "purple"
        ],
        "manufacturer" : "Hero",
        "performance" : 8,
        "timestamp" : ISODate("2023-01-10T00:00:00Z"),
        "price" : 72000
}
{
        "_id" : ObjectId("69b0eb708f1c10df3acef7e2"),
        "bike_name" : "Activa",
        "model" : "gearless",
        "category" : "125cc",
        "colors_available" : [
                "white",
                "red"
        ],
        "manufacturer" : "Honda",
        "performance" : 9,
        "timestamp" : ISODate("2023-05-15T00:00:00Z"),
        "price" : 85000
}
{
        "_id" : ObjectId("69b0eb708f1c10df3acef7e3"),
        "bike_name" : "Pulsar",
        "model" : "gear",
        "category" : "150cc",
        "colors_available" : [
                "black",
                "sport red"
        ],
        "manufacturer" : "Bajaj",
        "performance" : 7,
        "timestamp" : ISODate("2022-11-20T00:00:00Z"),
        "price" : 115000
}
```

```
{
        "_id" : ObjectId("69b0eb708f1c10df3acef7e4"),
        "bike_name" : "Duke",
        "model" : "gear",
        "category" : "200cc",
        "colors_available" : [
                "orange",
                "white"
        ],
        "manufacturer" : "KTM",
        "performance" : 10,
        "timestamp" : ISODate("2024-02-01T00:00:00Z"),
        "price" : 190000
}
{
        "_id" : ObjectId("69b0eb708f1c10df3acef7e5"),
        "bike_name" : "Jupiter",
        "model" : "gearless",
        "category" : "110cc",
        "colors_available" : [
                "blue",
                "grey"
        ],
        "manufacturer" : "TVS",
        "performance" : 4,
        "timestamp" : ISODate("2023-08-12T00:00:00Z"),
        "price" : 78000
}
```

```
> db.four_wheelers.find().pretty()
{
        "_id" : ObjectId("69b0ec2f8f1c10df3acef7f0"),
        "vehicle_name" : "Swift",
        "model" : "own",
        "category" : "car",
        "variants" : [
                "vxi",
                "zxi",
                "petrol"
        ],
        "manufacturer" : "Maruti",
        "performance" : 8,
        "timestamp" : ISODate("2023-03-10T00:00:00Z"),
        "price" : 700000
}
{
        "_id" : ObjectId("69b0ec2f8f1c10df3acef7f1"),
        "vehicle_name" : "City",
        "model" : "own",
        "category" : "car",
        "variants" : [
                "petrol",
                "diesel"
        ],
        "manufacturer" : "Honda",
        "performance" : 9,
        "timestamp" : ISODate("2022-09-15T00:00:00Z"),
        "price" : 1300000
}
{
        "_id" : ObjectId("69b0ec2f8f1c10df3acef7f2"),
        "vehicle_name" : "Eicher 10.5",
        "model" : "commercial",
        "category" : "mini truck",
        "variants" : [
                "diesel"
        ],
        "manufacturer" : "Eicher",
        "performance" : 7,
        "timestamp" : ISODate("2021-05-10T00:00:00Z"),
        "price" : 1500000
}
```

```
{
        "_id" : ObjectId("69b0ec758f1c10df3acef7f8"),
        "vehicle_name" : "Marcopolo",
        "model" : "commercial",
        "category" : "bus",
        "variants" : [
                "diesel"
        ],
        "manufacturer" : "Tata",
        "performance" : 4,
        "timestamp" : ISODate("2022-12-01T00:00:00Z"),
        "price" : 2800000
}
{
        "_id" : ObjectId("69b0ec758f1c10df3acef7f9"),
        "vehicle_name" : "Thar",
        "model" : "own",
        "category" : "car",
        "variants" : [
                "lx",
                "diesel",
                "petrol"
        ],
        "manufacturer" : "Mahindra",
        "performance" : 9,
        "timestamp" : ISODate("2023-06-25T00:00:00Z"),
        "price" : 1600000
}
```

7. Write a MongoDB query to display only vehicle name and price in all the collection of the database

```
> db.two_wheelers.find({}, { bike_name: 1, price: 1, _id: 0 })
{ "bike_name" : "Splendor", "price" : 72000 }
{ "bike_name" : "Activa", "price" : 85000 }
{ "bike_name" : "Pulsar", "price" : 115000 }
{ "bike_name" : "Duke", "price" : 190000 }
{ "bike_name" : "Jupiter", "price" : 78000 }
```

```
> db.four_wheelers.find({}, { vehicle_name: 1, price: 1, _id: 0 })
{ "vehicle_name" : "Swift", "price" : 700000 }
{ "vehicle_name" : "City", "price" : 1300000 }
{ "vehicle_name" : "Eicher 10.5", "price" : 1500000 }
{ "vehicle_name" : "Marcopolo", "price" : 2800000 }
{ "vehicle_name" : "Thar", "price" : 1600000 }
```

8. Write a MongoDB query to display two_wheelers from a particular company

```
> db.two_wheelers.find({ manufacturer: "Honda" })
{ "_id" : ObjectId("69b0eb708f1c10df3acef7e2"), "bike_name" : "Activa", "model" : "
gearless", "category" : "125cc", "colors_available" : [ "white", "red" ], "manufact
urer" : "Honda", "performance" : 9, "timestamp" : ISODate("2023-05-15T00:00:00Z"),
"price" : 85000 }
```

9. Write a MongoDB query to display four_wheelers available in diesel variants

```
> db.four_wheelers.find({ variants: "diesel" })
{ "_id" : ObjectId("69b0ec2f8f1c10df3acef7f1"), "vehicle_name" : "City", "model" :
"own", "category" : "car", "variants" : [ "petrol", "diesel" ], "manufacturer" : "H
onda", "performance" : 9, "timestamp" : ISODate("2022-09-15T00:00:00Z"), "price" :
1300000 }
{ "_id" : ObjectId("69b0ec2f8f1c10df3acef7f2"), "vehicle_name" : "Eicher 10.5", "mo
del" : "commercial", "category" : "mini truck", "variants" : [ "diesel" ], "manufac
turer" : "Eicher", "performance" : 7, "timestamp" : ISODate("2021-05-10T00:00:00Z")
, "price" : 1500000 }
{ "_id" : ObjectId("69b0ec2f8f1c10df3acef7f3"), "vehicle_name" : "Marcopolo", "mode
l" : "commercial", "category" : "bus", "variants" : [ "diesel" ], "manufacturer" :
"Tata", "performance" : 4, "timestamp" : ISODate("2022-12-01T00:00:00Z"), "price" :
 2800000 }
{ "_id" : ObjectId("69b0ec2f8f1c10df3acef7f4"), "vehicle_name" : "Thar", "model" :
"own", "category" : "car", "variants" : [ "lx", "diesel", "petrol" ], "manufacturer
" : "Mahindra", "performance" : 9, "timestamp" : ISODate("2023-06-25T00:00:00Z"), "
price" : 1600000 }
```

10. Write a MongoDB query to display vehicles name, category and manufacturer details whose rating is more than 5.

```
> db.two_wheelers.find({ performance: { $gt: 5 } }, { bike_name: 1, category: 1, ma
nufacturer: 1, _id: 0 }) performance: { $gt: 5 } }, { bike_name: 1, category: 1, ma
{ "bike_name" : "Splendor", "category" : "100cc", "manufacturer" : "Hero" }
{ "bike_name" : "Activa", "category" : "125cc", "manufacturer" : "Honda" }
{ "bike_name" : "Pulsar", "category" : "150cc", "manufacturer" : "Bajaj" }
{ "bike_name" : "Duke", "category" : "200cc", "manufacturer" : "KTM" }
```

```
{ db.four_wheelers.find({ performance: { $gt: 5 } }, { vehicle_name: 1, category: 1
, manufacturer: 1, _id: 0 })rformance: { $gt: 5 } }, { vehicle_name: 1, category: 1
{ "vehicle_name" : "Swift", "category" : "car", "manufacturer" : "Maruti" }
{ "vehicle_name" : "City", "category" : "car", "manufacturer" : "Honda" }
{ "vehicle_name" : "Eicher 10.5", "category" : "mini truck", "manufacturer" : "Eich
er" }
{ "vehicle_name" : "Thar", "category" : "car", "manufacturer" : "Mahindra" }
```

QUESTION 2 :

1. Create a database called ‘animal’ and write a MongoDB query to select database as ‘Animal’.

```
> use animal
switched to db animal
>
```

2. Write a MongoDB query to display all the databases.

```
> show dbs
admin     0.000GB
config    0.000GB
local     0.000GB
vehicles  0.000GB
```

3. Create a collection called ‘wild_animals’.(use capping) and Create a collection called ‘Domestic_animals’.

```
> db.createCollection("wild_animals", { capped: true, size: 5242880, max: 5000 })
{ "ok" : 1 }
> db.createCollection("domestic_animals")
{ "ok" : 1 }
>
```

4. Add 5 wild_animal details to the collection named ‘wild_animals’. Each document consists of following fields as animal_name, nature (harm or harmless), favorite_foods (meat, rabbits, deer etc) as array, care_taker_name, life span (in years), timestamp (when the animal registered at the Zoo) and expenses.

```
>   { animal_name: "Lion", nature: "harm", favorite_foods: ["meat", "deer"], care_taker  { animal_name: "Tig
er", nature: "harm", favorite_foods: ["meat", "wild boar"], ca  { animal_name: "Elephant", nature: "harmless
", favorite_foods: ["sugar cane", "ba  { animal_name: "Wolf", nature: "harm", favorite_foods: ["rabbits", "d
eer"], care_  { animal_name: "Giraffe", nature: "harmless", favorite_foods: ["leaves", "twigs"], care_taker_
name: "John", life_span: 25, timestamp: new Date("2019-09-12"), expenses: 4000 }er_name: "John", life_span:
25, timestamp: new Date("2019-09-12"), ex
...
... ])
{
        "acknowledged" : true,
        "insertedIds" : [
                ObjectId("69b0ef788f1c10df3acef7ff"),
                ObjectId("69b0ef788f1c10df3acef800"),
                ObjectId("69b0ef788f1c10df3acef801"),
                ObjectId("69b0ef788f1c10df3acef802"),
                ObjectId("69b0ef788f1c10df3acef803")
        ]
}
```

5. Add 5 domestic-animal details to the collection named ‘domestic_animals’. Each document consists of following fields as animal_name, gender (male or female), favorite_foods (meat, rabbits, deer etc) as array, animal_petname, life span (in years), timestamp (when the animal registered at the Zoo) and expenses.

```
> db  { animal_name: "Cow", gender: "female", favorite_foods: ["grass", "hay"], animal_petname: "Gauri", lif
e_sp  { animal_name: "Goat", gender: "male", favorite_foods: ["leaves", "grains"], animal_petname: "Billy",
life  { animal_name: "Rabbit", gender: "female", favorite_foods: ["carrots", "grass"], animal_petname: "Bunn
y",   { animal_name: "Sheep", gender: "male", favorite_foods: ["grass"], animal_petname: "Shaun", life_span:
 12,  { animal_name: "Horse", gender: "female", favorite_foods: ["oats", "grass"], animal_petname: "Starligh
t", life_span: 30, timestamp: new Date("2021-05-25"), expenses: 6000 }", "grass"], animal_petname: "Starligh
... ])
{
        "acknowledged" : true,
        "insertedIds" : [
                ObjectId("69b0ef9e8f1c10df3acef804"),
                ObjectId("69b0ef9e8f1c10df3acef805"),
                ObjectId("69b0ef9e8f1c10df3acef806"),
                ObjectId("69b0ef9e8f1c10df3acef807"),
                ObjectId("69b0ef9e8f1c10df3acef808")
        ]
}
```

6. Write a MongoDB query to display all documents available in wild_animals and Domastic_animals.

```
> db.wild_animals.find().pretty()
{
        "_id" : ObjectId("69b0ef788f1c10df3acef7ff"),
        "animal_name" : "Lion",
        "nature" : "harm",
        "favorite_foods" : [
                "meat",
                "deer"
        ],
        "care_taker_name" : "Robert",
        "life_span" : 15,
        "timestamp" : ISODate("2020-05-10T00:00:00Z"),
        "expenses" : 5000
}
{
        "_id" : ObjectId("69b0ef788f1c10df3acef800"),
        "animal_name" : "Tiger",
        "nature" : "harm",
        "favorite_foods" : [
                "meat",
                "wild boar"
        ],
        "care_taker_name" : "Robert",
        "life_span" : 12,
        "timestamp" : ISODate("2021-02-15T00:00:00Z"),
        "expenses" : 5500
}
{
        "_id" : ObjectId("69b0ef788f1c10df3acef801"),
        "animal_name" : "Elephant",
        "nature" : "harmless",
        "favorite_foods" : [
                "sugar cane",
                "banana"
        ],
        "care_taker_name" : "John",
        "life_span" : 60,
        "timestamp" : ISODate("2018-11-20T00:00:00Z"),
        "expenses" : 8000
}
{
        "_id" : ObjectId("69b0ef788f1c10df3acef802"),
        "animal_name" : "Wolf",
        "nature" : "harm",
        "favorite_foods" : [
                "rabbits",
                "deer"
        ],
        "care_taker_name" : "Alice",
        "life_span" : 8,
        "timestamp" : ISODate("2022-06-05T00:00:00Z"),
        "expenses" : 3000
}
{
        "_id" : ObjectId("69b0ef788f1c10df3acef803"),
        "animal_name" : "Giraffe",
        "nature" : "harmless",
        "favorite_foods" : [
                "leaves",
                "twigs"
        ],
        "care_taker_name" : "John",
        "life_span" : 25,
        "timestamp" : ISODate("2019-09-12T00:00:00Z"),
        "expenses" : 4000
}
```

```
> db.domestic_animals.find().pretty()
{
        "_id" : ObjectId("69b0ef9e8f1c10df3acef804"),
        "animal_name" : "Cow",
        "gender" : "female",
        "favorite_foods" : [
                "grass",
                "hay"
        ],
        "animal_petname" : "Gauri",
        "life_span" : 20,
        "timestamp" : ISODate("2023-01-01T00:00:00Z"),
        "expenses" : 2000
}
{
        "_id" : ObjectId("69b0ef9e8f1c10df3acef805"),
        "animal_name" : "Goat",
        "gender" : "male",
        "favorite_foods" : [
                "leaves",
                "grains"
        ],
        "animal_petname" : "Billy",
        "life_span" : 12,
        "timestamp" : ISODate("2023-03-15T00:00:00Z"),
        "expenses" : 1000
}
{
        "_id" : ObjectId("69b0ef9e8f1c10df3acef806"),
        "animal_name" : "Rabbit",
        "gender" : "female",
        "favorite_foods" : [
                "carrots",
                "grass"
        ],
        "animal_petname" : "Bunny",
        "life_span" : 8,
        "timestamp" : ISODate("2024-01-10T00:00:00Z"),
        "expenses" : 500
}
{
        "_id" : ObjectId("69b0ef9e8f1c10df3acef807"),
        "animal_name" : "Sheep",
        "gender" : "male",
        "favorite_foods" : [
                "grass"
        ],
        "animal_petname" : "Shaun",
        "life_span" : 12,
        "timestamp" : ISODate("2022-10-20T00:00:00Z"),
        "expenses" : 1200
}
{
        "_id" : ObjectId("69b0ef9e8f1c10df3acef808"),
        "animal_name" : "Horse",
        "gender" : "female",
        "favorite_foods" : [
                "oats",
                "grass"
        ],
        "animal_petname" : "Starlight",
        "life_span" : 30,
        "timestamp" : ISODate("2021-05-25T00:00:00Z"),
        "expenses" : 6000
}
> |
```

7. Write a MongoDB query to display only animal name and expenses in all the collection of the database

```
> db.wild_animals.find({}, { animal_name: 1, expenses: 1, _id: 0 })
{ "animal_name" : "Lion", "expenses" : 5000 }
{ "animal_name" : "Tiger", "expenses" : 5500 }
{ "animal_name" : "Elephant", "expenses" : 8000 }
{ "animal_name" : "Wolf", "expenses" : 3000 }
{ "animal_name" : "Giraffe", "expenses" : 4000 }
> db.domestic_animals.find({}, { animal_name: 1, expenses: 1, _id: 0 })
{ "animal_name" : "Cow", "expenses" : 2000 }
{ "animal_name" : "Goat", "expenses" : 1000 }
{ "animal_name" : "Rabbit", "expenses" : 500 }
{ "animal_name" : "Sheep", "expenses" : 1200 }
{ "animal_name" : "Horse", "expenses" : 6000 }
> |
```

8. Write a MongoDB query to display domestic_animals whose life is a particular year

```
> db.domestic_animals.find({ life_span: 12 })
{ "_id" : ObjectId("69b0ef9e8f1c10df3acef805"), "animal_name" : "Goat", "gender" : "male", "favorite_foods" : [ "leaves", "grains" ]
, "animal_petname" : "Billy", "life_span" : 12, "timestamp" : ISODate("2023-03-15T00:00:00Z"), "expenses" : 1000 }
{ "_id" : ObjectId("69b0ef9e8f1c10df3acef807"), "animal_name" : "Sheep", "gender" : "male", "favorite_foods" : [ "grass" ], "animal_
petname" : "Shaun", "life_span" : 12, "timestamp" : ISODate("2022-10-20T00:00:00Z"), "expenses" : 1200 }
> |
```

9. Write a MongoDB query to display wild_animals available under a particular Care_taker

```
> db.wild_animals.find({ care_taker_name: "Robert" })
{ "_id" : ObjectId("69b0ef788f1c10df3acef7ff"), "animal_name" : "Lion", "nature" : "harm", "favorite_foods" : [ "meat", "deer" ], "c
are_taker_name" : "Robert", "life_span" : 15, "timestamp" : ISODate("2020-05-10T00:00:00Z"), "expenses" : 5000 }
{ "_id" : ObjectId("69b0ef788f1c10df3acef800"), "animal_name" : "Tiger", "nature" : "harm", "favorite_foods" : [ "meat", "wild boar"
 ], "care_taker_name" : "Robert", "life_span" : 12, "timestamp" : ISODate("2021-02-15T00:00:00Z"), "expenses" : 5500 }
> |
```

10. Write a MongoDB query to display animal name, favorite_foods and expenses details whose lifespan is more than 5 years.

```
> db.wild_animals.find({ life_span: { $gt: 5 } }, { animal_name: 1, favorite_foods: 1, expenses: 1, _id: 0 })
{ "animal_name" : "Lion", "favorite_foods" : [ "meat", "deer" ], "expenses" : 5000 }
{ "animal_name" : "Tiger", "favorite_foods" : [ "meat", "wild boar" ], "expenses" : 5500 }
{ "animal_name" : "Elephant", "favorite_foods" : [ "sugar cane", "banana" ], "expenses" : 8000 }
{ "animal_name" : "Wolf", "favorite_foods" : [ "rabbits", "deer" ], "expenses" : 3000 }
{ "animal_name" : "Giraffe", "favorite_foods" : [ "leaves", "twigs" ], "expenses" : 4000 }
> db.domestic_animals.find({ life_span: { $gt: 5 } }, { animal_name: 1, favorite_foods: 1, expenses: 1, _id: 0 })
{ "animal_name" : "Cow", "favorite_foods" : [ "grass", "hay" ], "expenses" : 2000 }
{ "animal_name" : "Goat", "favorite_foods" : [ "leaves", "grains" ], "expenses" : 1000 }
{ "animal_name" : "Rabbit", "favorite_foods" : [ "carrots", "grass" ], "expenses" : 500 }
{ "animal_name" : "Sheep", "favorite_foods" : [ "grass" ], "expenses" : 1200 }
{ "animal_name" : "Horse", "favorite_foods" : [ "oats", "grass" ], "expenses" : 6000 }
> |
```